

Fairness Experiment Report - Upload 52

Fairness Experiment Report - Upload 52 summarizes the current fairness state for upload 52. Accuracy changed by -0.015 and fairness score changed by +0.099. Good fairness gain with acceptable accuracy trade-off. The best balanced model is ThresholdOptimizer_mitigated_threshold with combined score 0.909.

Upload ID: 52

Dataset: adult.csv

Model: adult_income_pipeline.pkl

Best Balanced Model: ThresholdOptimizer_mitigated_threshold

Best Accuracy Model: Pipeline_mitigated_reweighting

Best Fairness Model: ThresholdOptimizer_mitigated_threshold

Accuracy: 0.838

Fairness Score: 0.955

DPD: 0.089

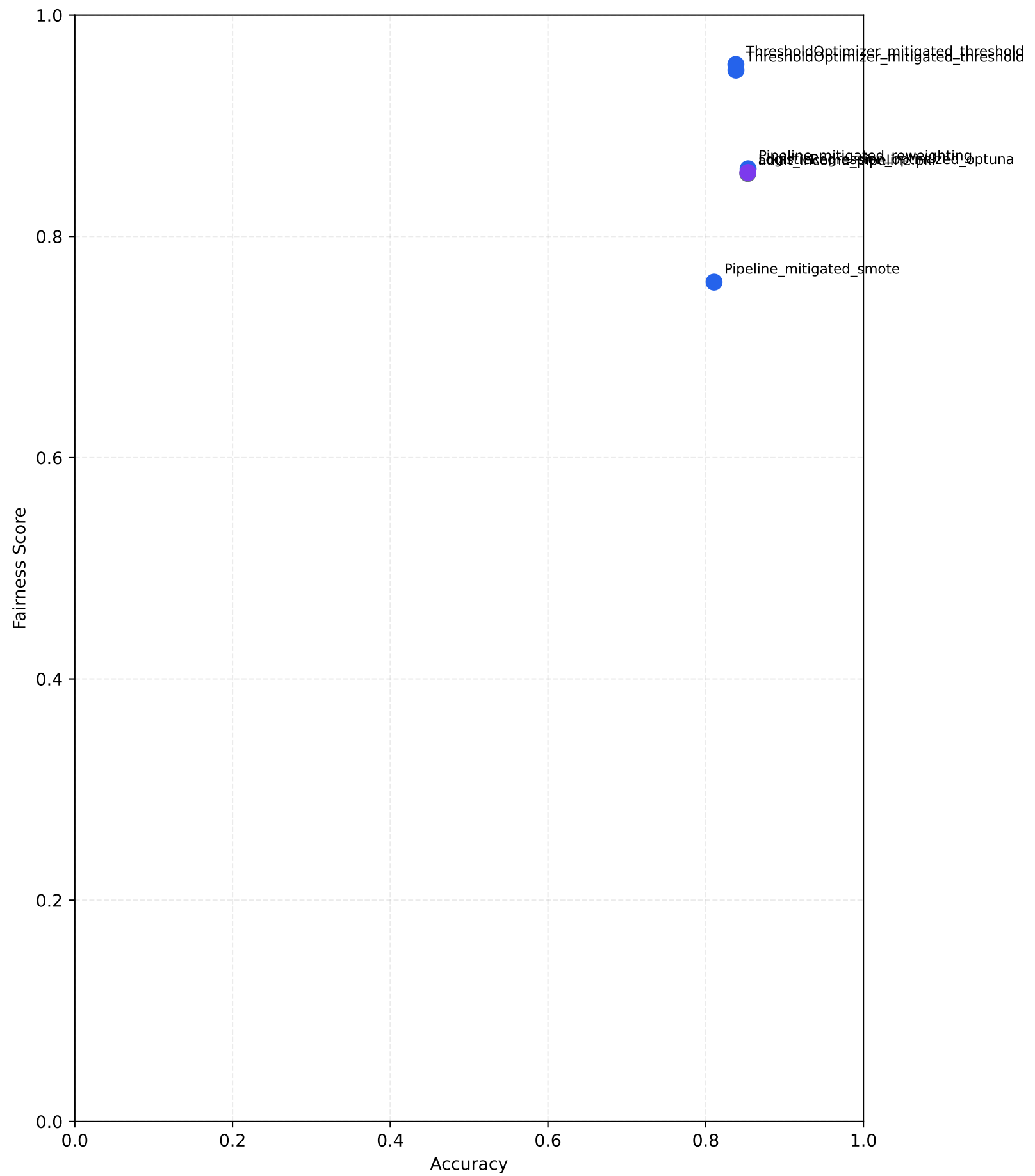
EOD: 0.001

DIR: 0.551

Sections: comparison, optimization

Generated at: 2026-05-13T06:14:19.759918Z

Accuracy vs Fairness Trade-off



Model Ranking and Interpretation

Good fairness gain with acceptable accuracy trade-off. The best balanced model is ThresholdOptimizer_mitigated_threshold with combined score 0.909.

Mitigated: ThresholdOptimizer_mitigated_threshold | accuracy=0.838, fairness=0.955, combined=0.909

Mitigated: ThresholdOptimizer_mitigated_threshold | accuracy=0.838, fairness=0.950, combined=0.905

Mitigated: Pipeline_mitigated_reweighting | accuracy=0.854, fairness=0.861, combined=0.858

Original: adult_income_pipeline.pkl | accuracy=0.853, fairness=0.857, combined=0.855

Optimized: LogisticRegression_optimized_optuna | accuracy=0.853, fairness=0.858, combined=0.855

Mitigated: Pipeline_mitigated_smote | accuracy=0.811, fairness=0.759, combined=0.779